BASIC INTRODUCTION (1-2)

Networks or systems of trails form through a common positive-feedback mechanism across a range of domains from eusocial insects, to bear trails through the wilderness to human foot paths, to road networks.

MORE DETAILED BACKGROUND (2-3)

Active-walker models show that simple behavior rules can reproduce the tree-like networks of foraging insects, as well as qualitative features of emergent pedestrian paths through parks. In these models, walkers modify the environment as they move and use local gradient detection to move towards certain targets while preferring to walk over ground that's been frequently used.

GENERAL PROBLEM (1)

However, existing models may apply poorly where walkers have more complete information of their goals and the envirnoment, and can plan their behavior intelligently.

HERE WE SHOW (1)

Here we investigate a model in which agents take a least-cost path across a fully-revealed environment to their destination.

MAIN RESULT (2-3)

Our model reproduces the qualitative features of networks produced by gradient-based models under simple scenarios (small numbers of destinations, or CPT) while also producing much more intricate and sophisticated networks in more complex scenarios (larger numbers of destinations, or where destinations move or vary). Furthermore, because our model defines behavior in terms of costs we are able to quantify agent behavior and networks to show quantitatively that walkers allocate trail efficiently while obeying certain geometrical constraints.

GENERAL CONTEXT (1-2)

The tenets of our model are extremely simple: an environment parameterized by costs decreased by frequent use, across which agents intelligently minimize their travel costs. Therefore, our results suggest that across a wide range of settings, stigmergy produces transportation networks with common and quantifiable features and dynamics.